DMFR-YOLO: Defect Morphology Encoding and Feature Reassembly for Fabric Defect Detection

Kangzhao Tan[#], Jiarui Zhang[#], Junjie Yu, Heng Wu, Jing Lin[*], and Chunhua He[*]

***Abstract*—Circular knitting machines are widely used in continuous fabric production. Needle or yarn abnormalities can cause repeated defects along the fabric before they are detected, resulting in material waste and economic losses. To enable in-process defect detection and intervention, we develop a closed-loop defect detection system with edge inference and wired machine-stop control, and propose a lightweight detector termed DMFR-YOLO. DMFR-YOLO introduces defect morphology encoding (DME) into the backbone to explicitly model horizontally elongated defects, vertically elongated defects, and localized hole-like defects. In the feature pyramid, content-adaptive feature reassembly (CAFR) replaces fixed upsampling with content-adaptive reconstruction, improving the preservation of defect boundaries and weak responses during multi-scale feature fusion. Evaluated on a self-collected fabric defect dataset from a real-world circular knitting machine production line, DMFR-YOLO achieves 66.3% AP and 96.9% AP50, improving the YOLOv8n baseline by 7.0 and 2.8 percentage points, respectively, with only 3.39 M parameters. The detector is deployed as an INT8 RKNN model on an RK3566 edge device. The resulting system couples real-time defect detection, automatic local stop control, operator review, and cloud-based quality analysis, providing a practical solution for reducing defect propagation during circular-knitting production.**

***Index Terms*—Industrial defect detection, YOLO, deep learning, edge computing, Internet of Things.**

I. Introduction

CIRCULAR knitting machines are widely used in weft-knit fabric production and support high-rate, large-scale manufacturing [1], [2]. The resulting fabrics are used in everyday garments such as T-shirts and sportswear [2], [3]. The International Textile Manufacturers Federation reported global shipments of approximately 24,500 large circular knitting machines in 2025 [4]. Conventional quality checks are often performed after knitting is completed. Persistent needle or yarn abnormalities may therefore generate repeated defects along the fabric before detection, resulting in fabric damage and economic losses. [5]–[7]. This motivates in-process defect detection to enable timely intervention and limit defect propagation during continuous production.

Single-stage YOLO detectors combine rapid inference with explicit bounding-box localization [8], making them suitable for industrial defect detection. YOLO-based approaches have been applied to steel defects and large stamped parts [9]–[11]. For fabrics, existing designs mainly pursue lightweight architectures for efficient inference [12], [13], adaptive or deformable feature extraction for irregular defects [14], [15], and improved multi-scale fusion or attention for small and weak defects [16]–[21]. Together, these studies demonstrate the suitability of YOLO-based detectors for fabric defect detection. However, these improvements mainly focus on detector efficiency or general feature enhancement, with less attention paid to the specific spatial characteristics of fabric defects.

In circular-knitted fabrics, the three defect categories considered in this work exhibit distinct spatial morphologies: horizontally elongated defects, vertically elongated defects, and localized hole-like defects. Conventional feature extraction can learn these patterns, but it does not explicitly encode their spatial characteristics. In addition, nearest-neighbor upsampling enlarges feature maps through fixed replication without adapting to local feature content. This may limit the preservation of thin defect boundaries and weak responses during multi-scale feature fusion.

For production-line operation, defect detection must be coordinated with the machine and support local intervention. We implement a hardware and detection system comprising an industrial camera with LED illumination, a Hall sensor, and an RK3566 edge device. Machine rotation detected by the Hall sensor activates local detection. When a defect is detected, the system issues a wired stop command and displays the detection result for operator review. The IoT cloud receives defect and device records for monitoring and production-quality analysis, but does not participate in the stop decision [22].

Within this system, we develop DMFR-YOLO by modifying feature encoding and feature reconstruction. Defect morphology encoding (DME) introduces horizontal, vertical, and local branches at selected backbone stages to strengthen the representation of the three characteristic defect morphologies. Content-adaptive feature reassembly (CAFR) predicts reassembly weights from local feature content and replaces fixed top-down feature-pyramid upsampling with location-adaptive neighborhood aggregation. Together, the two modules encode the spatial forms of defects in the backbone and adaptively reconstruct these features across pyramid scales.

The main contributions of our work are as follows.

(1) We design and implement a production-line defect detection system integrating an industrial camera with LED illumination, Hall-sensor-based machine monitoring, RK3566 edge inference, wired machine-stop control, operator review, and cloud-based monitoring, and demonstrate its deployment on a real circular-knitting production line.

(2) We introduce DME to encode the characteristic morphologies of horizontal defects, vertical defects, and holes through horizontal, vertical, and local branches,

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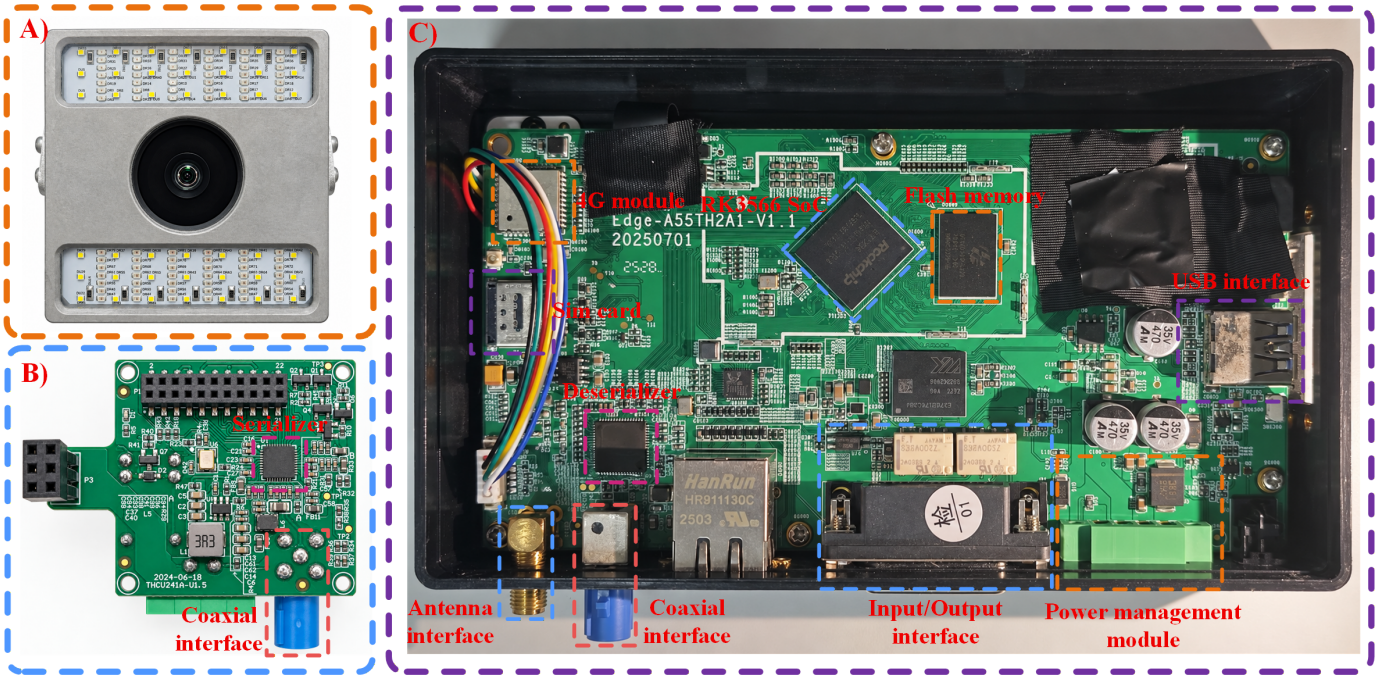

Fig. 1. Hardware of the production-line defect detection system: (A) industrial camera with LED illumination; (B) camera-side interface board with a serializer and coaxial connector; and (C) RK3566 edge board with the labeled communication, memory, camera-interface, input/output, and power-management components.

providing a lightweight morphology-aware representation for circular-knitted defect detection.

(3) We incorporate CAFR into the feature pyramid to replace fixed upsampling with locally predicted reassembly weights, enabling content-adaptive reconstruction during multi-scale fusion.

II. Hardware and Detection System

A. Hardware Configuration

The system combines image acquisition, machine-state sensing, edge computation, and a wired connection to the knitting-machine controller. Fig. 1 shows the camera assembly and edge-device hardware. The industrial camera and LED light are integrated in the front assembly (A). A camera-side interface board (B) provides a serializer and a coaxial connector, while the RK3566 edge board (C) includes a deserializer for the camera connection. This arrangement connects the camera mounted near the fabric to the edge device responsible for detection and machine control.

The RK3566 edge board integrates flash memory, power management, and input/output interfaces, together with a 4G module, SIM-card slot, and antenna connector for cloud communication. The Hall sensor provides the machine-rotation signal, and a wired connection to the machine controller carries the stop signal. Image processing and the stop decision remain on the edge device; cloud communication is used for device monitoring and defect records rather than for latency-critical control.

B. Closed-Loop Detection Workflow

Fig. 2 shows the installation of our defect detection system on a circular knitting machine production line. After the operator starts the circular knitting machine, the Hall sensor detects rotation and signals the RK3566 edge device to switch on the LED light and begin detection from the camera video stream. When a defect is detected, the device sends a stop signal to the machine controller through a wired connection to interrupt further defect propagation. The defect image and detection result are displayed locally for operator review and troubleshooting. The operator resolves the problem and restarts the machine, completing the operational cycle.

C. Edge Deployment and Cloud Platform

The detector described in Section III is deployed locally on the RK3566 edge device. To enable on-device inference, the trained model is converted from PyTorch to ONNX and then to an INT8 RKNN model. The trained model is exported with a fixed 640×640 input and converted using RKNN-Toolkit2 [23], with post-training INT8 quantization for efficient NPU execution [24]. An Android application executes the RKNN model on the RK3566 edge device.

The cloud platform receives defect records and device states for centralized monitoring, defect statistics, and production-quality analysis. Its interface supports device-state visualization, remote configuration, and review of uploaded defect images (Fig. 3).

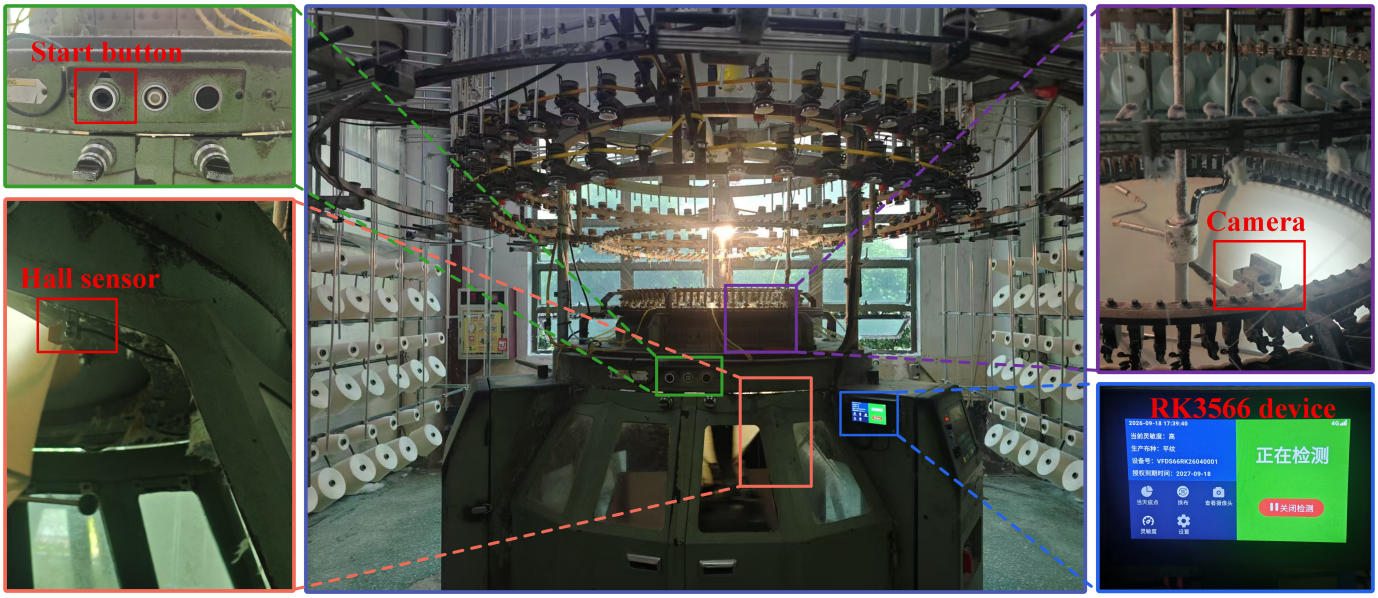

Fig. 2. Actual deployment on the circular knitting machine. The insets highlight the start button, Hall sensor, industrial camera, and RK3566 edge device, respectively.

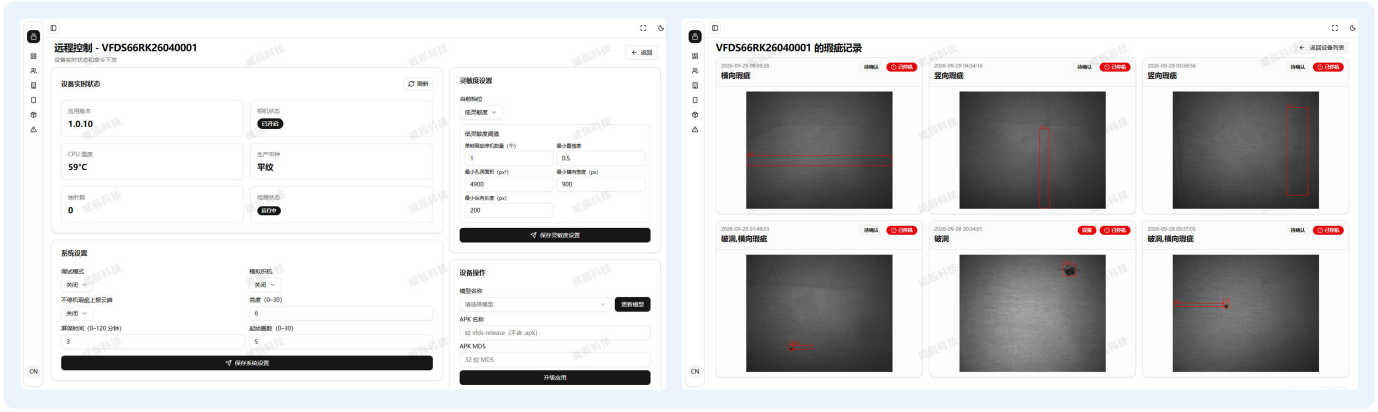

Fig. 3. IoT cloud interface: device-state monitoring and remote configuration (left), and uploaded defect records (right). Cloud functions support centralized supervision and quality analysis.

III. Proposed Method

A. DMFR-YOLO Architecture

DMFR-YOLO is developed based on YOLOv8n, which is selected for its compatibility with RK3566 deployment, compact model size, and modular architecture [25], [26]. The proposed modifications focus on defect morphology encoding in the backbone and feature reconstruction in the feature pyramid.

Fig. 4 shows the overall architecture. DME is introduced after the main backbone stages to enhance the representation of different defect morphologies. The extracted multi-scale features are then passed to the neck for feature fusion [27], [28]. Within the feature pyramid, CAFR replaces two nearest-neighbor upsampling operations with content-adaptive feature reassembly to better preserve defect boundaries and weak responses. The fused features are subsequently used by the multi-scale detection head to predict defect locations and categories.

B. Defect Morphology Encoding

The three defect categories considered in this work present different spatial characteristics. Horizontal and vertical defects usually extend along one dominant direction, whereas hole defects are more localized. To capture these differences, we introduce a defect morphology encoding (DME) module with three parallel branches, as shown in the enlarged view in Fig. 4. The horizontal and vertical branches use 1×7 and 7×1 depthwise convolutions to enlarge the receptive field along the corresponding spatial directions, while the 3×3 branch captures localized defect structures. Depthwise convolution is adopted to model these spatial patterns with low computational overhead before channel-wise feature fusion [29].

For a C2f output $X \in \mathbb{R}^{C \times H \times W}$, the three branch responses are

$$F_h = \mathrm{DW}_{1\times7}(X), \tag{1}$$

$$F_v = \mathrm{DW}_{7\times1}(X), \tag{2}$$

$$F_l = \mathrm{DW}_{3\times3}(X). \tag{3}$$

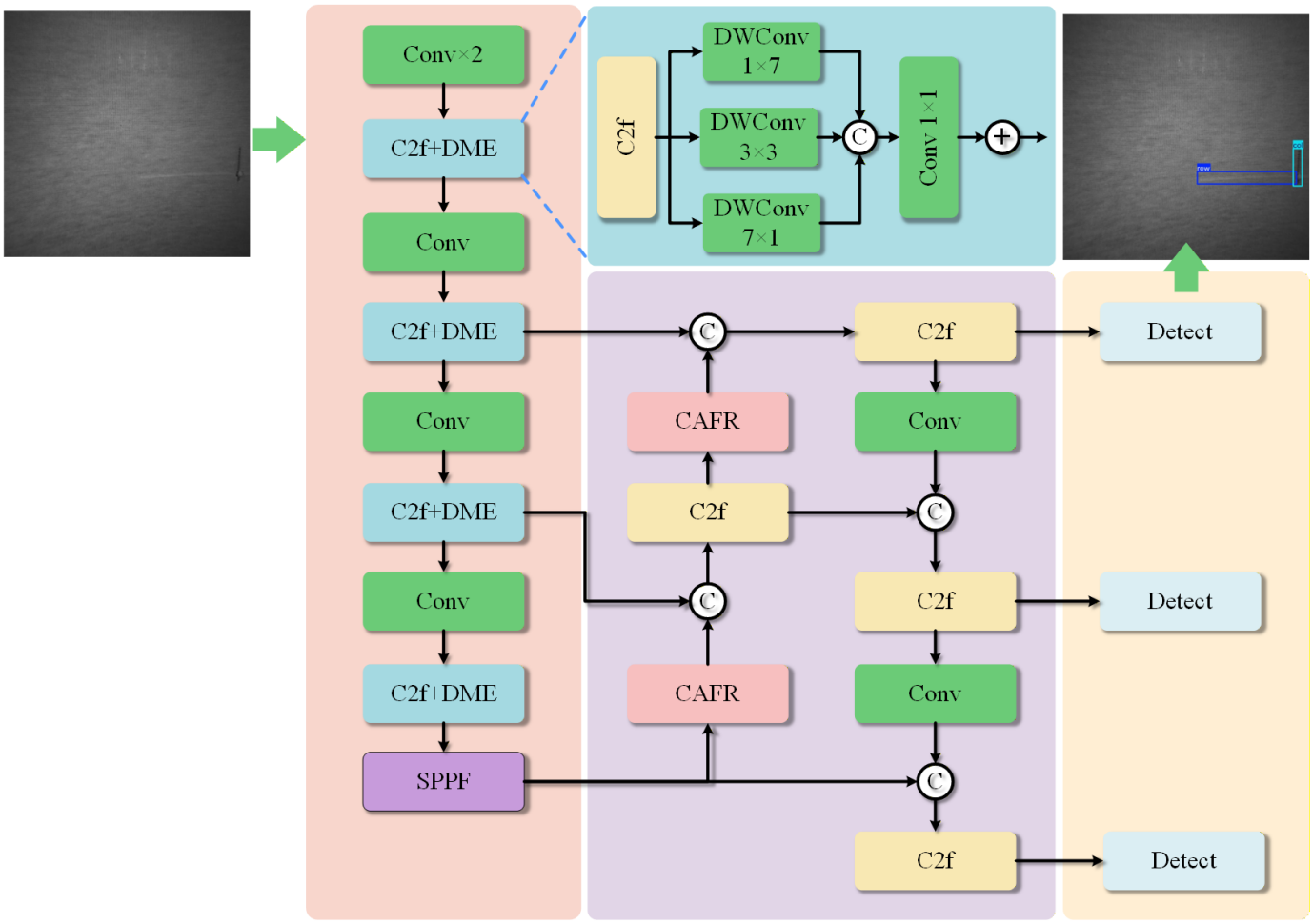

Fig. 4. Architecture of DMFR-YOLO with an enlarged view of DME. DME follows the backbone C2f stages, and CAFR replaces the two top-down upsampling operations. The neck fusion blocks, bottom-up path, and three-scale detection head are retained from YOLOv8n.

Here, DW denotes depthwise convolution followed by batch normalization and SiLU activation. All branches preserve the spatial dimensions and produce C output channels. Their concatenation is fused as

$$F_d = \phi\left([F_h, F_v, F_l]\right), \tag{4}$$

where $[\cdot]$ denotes channel concatenation and ϕ is a 1×1 convolution mapping $3C$ channels to C, followed by batch normalization and SiLU. This learnable fusion combines the three types of spatial evidence rather than assigning a fixed branch to a particular defect class. A residual connection [30] gives

$$Y = X + F_d. \tag{5}$$

The residual connection combines the morphology-aware features with the incoming C2f output, allowing DME to enhance the backbone representation without discarding the features learned by the preceding layers. The directional branches are based on asymmetric convolution designs [31].

C. Content-Adaptive Feature Reassembly

In the feature pyramid, nearest-neighbor upsampling enlarges feature maps by directly replicating feature values. This fixed operation does not adapt to the spatial structure of defects. To address this limitation, we introduce content-adaptive feature reassembly (CAFR), following the reassembly mechanism of Content-Aware ReAssembly of FEatures(CARAFE) [32]. CAFR predicts spatially varying kernels from the input features and uses them to aggregate local neighborhoods (Fig. 5), aiming to better preserve defect boundaries and weak responses during multi-scale fusion.

In DMFR-YOLO, CAFR replaces the upsampling operations at both top-down fusion stages, with an upsampling factor of $s = 2$ and a reassembly kernel size of $k = 5$. As shown in Fig. 5, CAFR consists of two parallel branches for kernel prediction and local feature extraction. Their outputs are subsequently combined through weighted aggregation to generate the upsampled features. The two branches and their subsequent feature reassembly are described as follows:

1) Reassembly Kernel Prediction: Given an input feature map $X \in \mathbb{R}^{C\times H\times W}$, a 1×1 convolution first reduces its channel dimension:

$$X_c = E_{1\times1}(X). \tag{6}$$

Here, $X_c \in \mathbb{R}^{D\times H\times W}$, where $D = \max(\lfloor C/4 \rfloor, 16)$ denotes the compressed channel count; $E_{1\times1}$ includes batch normal-

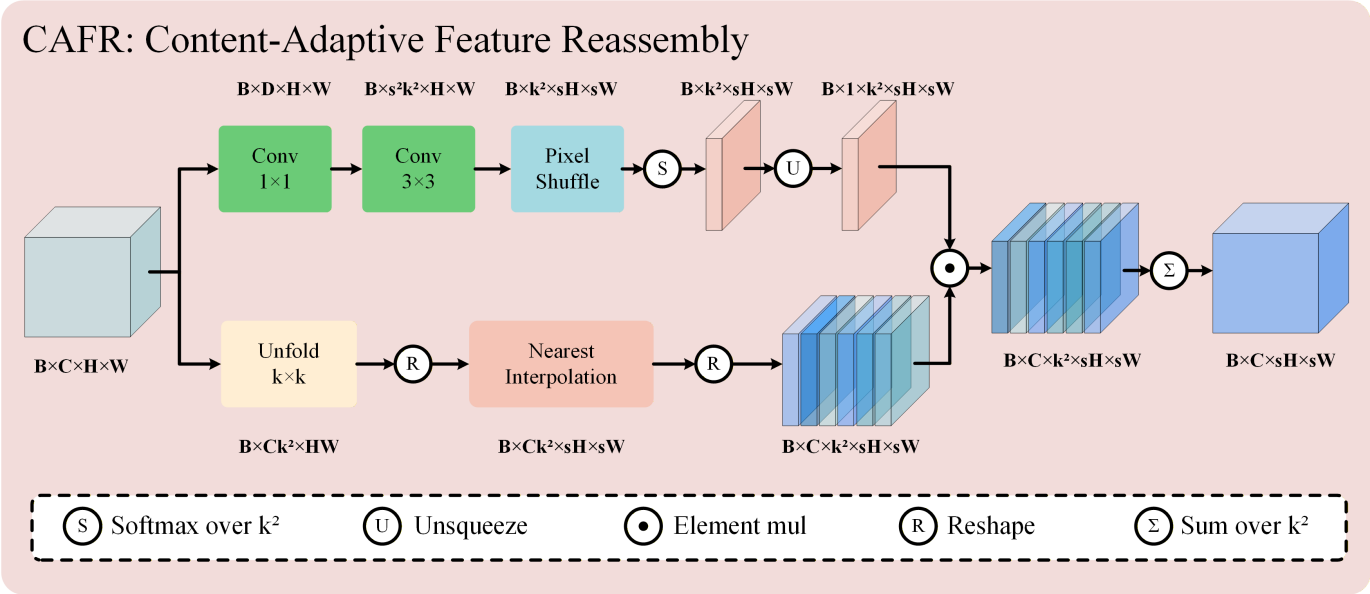

Fig. 5. CAFR structure. The upper path predicts location-specific 5×5 reassembly kernels, and the lower path unfolds local feature patches and replicates them to align with output locations. Weighted aggregation, rather than this alignment operation, produces the final $2\times$ upsampled feature.

ization and SiLU activation. A linear 3×3 convolution then predicts the unnormalized reassembly weights:

$$K' = E_{3\times3}(X_c). \tag{7}$$

The resulting tensor is $K' \in \mathbb{R}^{s^2k^2\times H\times W}$. For $s = 2$ and $k = 5$, this corresponds to 100 weight channels at each input location. Pixel Shuffle rearranges the predicted weights onto the higher-resolution grid, followed by Softmax normalization:

$$K = \mathrm{Softmax}(\mathrm{PS}(K')). \tag{8}$$

Softmax is applied along the k^2 neighborhood dimension. The resulting tensor $K \in \mathbb{R}^{k^2\times sH\times sW}$ provides a normalized reassembly kernel for each output location.

2) Local Feature Extraction: In parallel, the patch branch extracts local neighborhoods from the input feature map using Unfold. The extracted patches are reshaped and upsampled using nearest-neighbor interpolation to match the spatial resolution of the predicted reassembly kernels:

$$\begin{aligned} P' &= \mathrm{Reshape}(\mathrm{Unfold}_{k\times k}(X)), \\ P &= \mathrm{Reshape}(\mathrm{Nearest}_s(P')). \end{aligned} \tag{9}$$

Here, $\mathrm{Unfold}_{k\times k}$ extracts a $k \times k$ neighborhood around each input position, and $\mathrm{Nearest}_s$ denotes nearest-neighbor upsampling with scale factor $s = 2$. The first reshape produces $P' \in \mathbb{R}^{Ck^2\times H\times W}$; the final reshape gives the aligned patch tensor $P \in \mathbb{R}^{C\times k^2\times sH\times sW}$, where k^2 indexes the positions within each local neighborhood. Nearest-neighbor interpolation is used only for neighborhood alignment, not to generate the final output features.

3) Feature Reassembly: The predicted kernels and the aligned local features are combined through element-wise multiplication and summation along the neighborhood dimension:

$$Y = \sum_{u=1}^{k^2} K_u \odot P_u. \tag{10}$$

Here, $K_u \in \mathbb{R}^{sH\times sW}$ and $P_u \in \mathbb{R}^{C\times sH\times sW}$ denote the weight map and corresponding local feature map for the u-th neighborhood position, respectively. The symbol $\odot$ denotes element-wise multiplication, with the weights broadcast across feature channels, producing an output feature map $Y \in \mathbb{R}^{C\times sH\times sW}$.

By predicting location-specific reassembly weights, CAFR adaptively aggregates local features during upsampling, helping preserve defect-related information in multi-scale feature fusion.

IV. Experimental Setup

A. Dataset

Images were collected during operation of a real circular-knitting production line. A global-shutter industrial camera captures 1280×1024 frames approximately 10 cm from the fabric, with a field of view of approximately 16×14 cm. We selected images with visible defects and manually annotated their bounding boxes. The dataset includes three defect categories: horizontal defects (*row*), vertical defects (*col*), and holes (*hole*).

The dataset contains 403 images with 468 annotated defect instances, divided into training, validation, and test sets of 282, 40, and 81 images, respectively, as summarized in Table I. The validation set is used for checkpoint selection, while performance is reported on the test set containing 91 labeled defects.

TABLE I
IMAGE-LEVEL SPLIT AND DEFECT-INSTANCE COUNTS IN THE REAL PRODUCTION-LINE DATASET.

Split	Images	Instances	Row	Col	Hole
Train	282	329	174	75	80
Validation	40	48	24	13	11
Test	81	91	60	12	19

TABLE II
IMPLEMENTATION PROTOCOL. TRAINING SETTINGS DESCRIBE THE ULTRALYTICS EXPERIMENTS; DEPLOYMENT SETTINGS DESCRIBE THE INSTALLED EDGE DEVICE.

Category	Parameter	Specification
1. Training setup	Hardware	Intel i7-13700KF; NVIDIA RTX 3090 (24 GB)
	Software	Ubuntu 24.04, Python 3.10.20, PyTorch 2.4.1+cu124, and Ultralytics 8.4.140
2. Deployment setup	Hardware	Rockchip RK3566
	System	Android 13
	Quantization	INT8
3. Data protocol	Input resolution	Fixed at 640×640
4. Training dynamics	Batch	16
	Maximum epochs	300
	Early-stopping patience	80 epochs

B. Implementation Details

Table II summarizes the training and deployment settings. DMFR-YOLO and YOLOv8n are initialized using the pretrained yolov8n.pt weights, while newly added layers use default initialization. Other comparison models use the available pretrained checkpoints in their respective implementations. The remaining Ultralytics training settings follow the framework defaults.

The comparisons include lightweight YOLO models [25], [33], DETR [34], Deformable DETR [35], Faster R-CNN [36], and an R18 model from the RT-DETRv2 implementation [37], denoted RT-DETR-R18. All models are evaluated on the same test set using their respective frameworks and evaluation settings.

C. Evaluation Metrics

We report COCO average precision (AP) across ten IoU thresholds and AP at IoU 0.50 (AP50) [38]. For a predicted box B_p and a ground-truth box B_g, intersection over union is $\mathrm{IoU}(B_p, B_g) = |B_p \cap B_g|/|B_p \cup B_g|$. Let $\mathrm{AP}_c(t)$ denote the average precision for class c at IoU threshold t, and let N_c denote the number of defect categories. The metrics are defined as follows:

$$\mathrm{AP} = \frac{1}{10N_c} \sum_{c=1}^{N_c} \sum_{t \in T} \mathrm{AP}_c(t), \tag{11}$$

$$\mathrm{AP}_{50} = \frac{1}{N_c} \sum_{c=1}^{N_c} \mathrm{AP}_c(0.50), \tag{12}$$

where $T = \{0.50, 0.55, \ldots, 0.95\}$ and $N_c = 3$ in this study. All AP values are reported as percentages. AP therefore evaluates localization over increasingly strict overlap requirements, whereas AP50 uses a single, less restrictive threshold. Differences between reported AP values are expressed in percentage points.

Model parameters and FLOPs are measured using the respective frameworks with an input resolution of 640×640.

V. RESULTS AND DISCUSSION

A. Comparison With Other Detectors

Table III presents the detection results on the test set. DMFR-YOLO achieves 66.3% AP and 96.9% AP50, improving the YOLOv8n baseline by 7.0 and 2.8 percentage points, respectively. These improvements are achieved with only 0.38 M additional parameters and approximately 0.8 G additional FLOPs. The larger improvement in AP indicates better detection performance under stricter IoU thresholds.

Among the evaluated lightweight YOLO detectors, DMFR-YOLO achieves the highest AP. Although RT-DETR-R18 attains a slightly higher AP of 67.7%, it requires 20.09 M parameters and 60.72 G FLOPs, compared with 3.39 M parameters and 8.90 G FLOPs for DMFR-YOLO. Fig. 6 further illustrates the favorable balance between detection accuracy and model complexity achieved by DMFR-YOLO.

Fig. 7 compares the detection results of different models on seven selected test images, highlighting differences in detection accuracy and bounding-box localization.

B. Component Ablation

Table IV evaluates the individual and combined effects of DME and CAFR on detection performance. Introducing

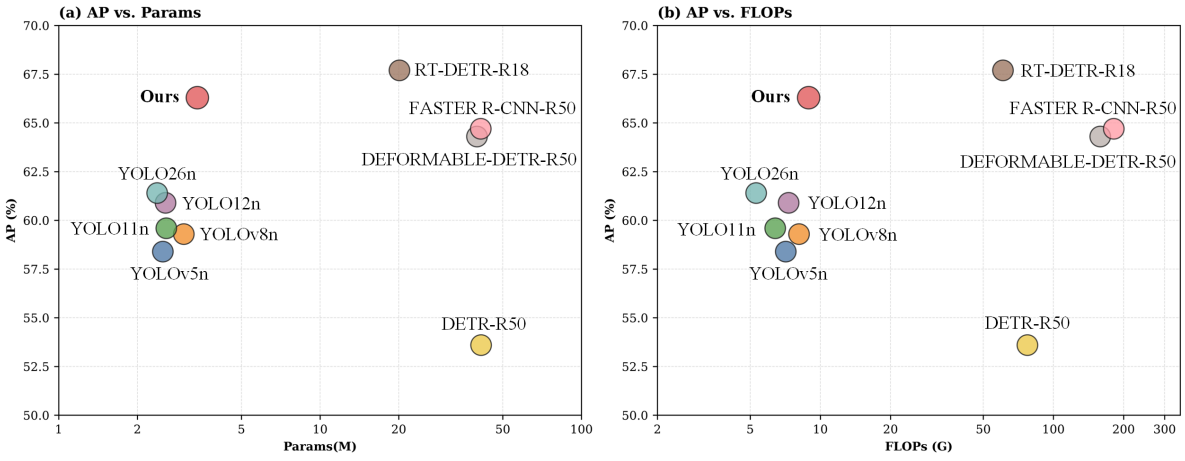

Fig. 6. AP versus Params (left) and FLOPs (right). Both horizontal axes are logarithmic. Ours denotes DMFR-YOLO.

TABLE III
DETECTION RESULTS ON THE TEST SET. BOLD AND UNDERLINED VALUES INDICATE THE HIGHEST AND SECOND-HIGHEST REPORTED SCORES, RESPECTIVELY.

Model	Params (M)	FLOPs (G)	AP50 (%)	AP (%)
Lightweight YOLO Detectors				
YOLOv5n	2.50	7.10	93.1	58.4
YOLO11n	2.58	6.40	95.0	59.6
YOLO12n	2.56	7.30	94.3	60.9
YOLO26n	2.38	5.30	96.5	61.4
Transformer Detectors				
RT-DETR-R18	20.09	60.72	95.2	**67.7**
DETR-R50	41.28	77.25	96.4	53.6
Deformable DETR-R50	39.82	158.66	<u>96.8</u>	64.3
Two-Stage Detector				
Faster R-CNN-R50	41.13	181.45	96.1	64.7
Baseline				
YOLOv8n	3.01	8.10	94.1	59.3
Ours				
DMFR-YOLO	3.39	8.90	**96.9**	<u>66.3</u>

TABLE IV
COMPONENT ABLATION. PARAMETERS ARE IN MILLIONS, FLOPS IN BILLIONS, AND AP50/AP IN PERCENT.

Variant	DME	CAFR	Params	FLOPs	AP50	AP
YOLOv8n	–	–	3.01	8.1	94.1	59.3
+ DME	✓	–	3.28	8.8	94.2	64.1
+ CAFR	–	✓	3.11	8.2	94.9	60.7
DMFR-YOLO	✓	✓	3.39	8.9	**96.9**	**66.3**

DME increases AP from 59.3% to 64.1%, while CAFR alone improves AP to 60.7%. When both modules are incorporated, DMFR-YOLO achieves 66.3% AP and 96.9% AP50, outperforming either individual modification. These results demonstrate the complementary effects of the two modules. DME enhances the representation of defect morphologies in the backbone, while CAFR adaptively reconstructs these features during multi-scale fusion. Their combination further improves the overall detection performance.

Fig. 8 compares the feature visualizations of YOLOv8n and DMFR-YOLO. Compared with YOLOv8n, DMFR-YOLO exhibits more concentrated feature responses around elongated defects and holes, with activation regions more closely aligned with the defect shapes. These observations suggest that DMFR-YOLO provides a more focused representation of defect regions.

VI. CONCLUSION

This paper presents DMFR-YOLO and a defect detection system for circular knitting machines. DMFR-YOLO introduces defect morphology encoding (DME) to capture the spatial characteristics of horizontal defects, vertical defects, and holes. Content-adaptive feature reassembly (CAFR) improves feature reconstruction during multi-scale fusion. Experiments on the production line dataset show that DMFR-YOLO achieves 66.3% AP and 96.9% AP50, outperforming YOLOv8n by 7.0 and 2.8 percentage points, respectively, with 3.39 M parameters and 8.9 G FLOPs. The ablation results further demonstrate the complementary effects of DME and CAFR.

The proposed detector is deployed on an RK3566 edge device and integrated with machine monitoring and wired stop control. The system enables automatic defect detection and timely intervention without relying on cloud communication. Detection results are displayed locally for operators, while the cloud platform supports equipment monitoring and production quality analysis. By integrating defect detection with machine control and quality management, this work provides a complete and practical solution for fabric defect detection in circular knitting production. Future work will explore the application of DMFR-YOLO to different knitted fabrics with varying texture patterns and defect characteristics to evaluate its generalization across production conditions.

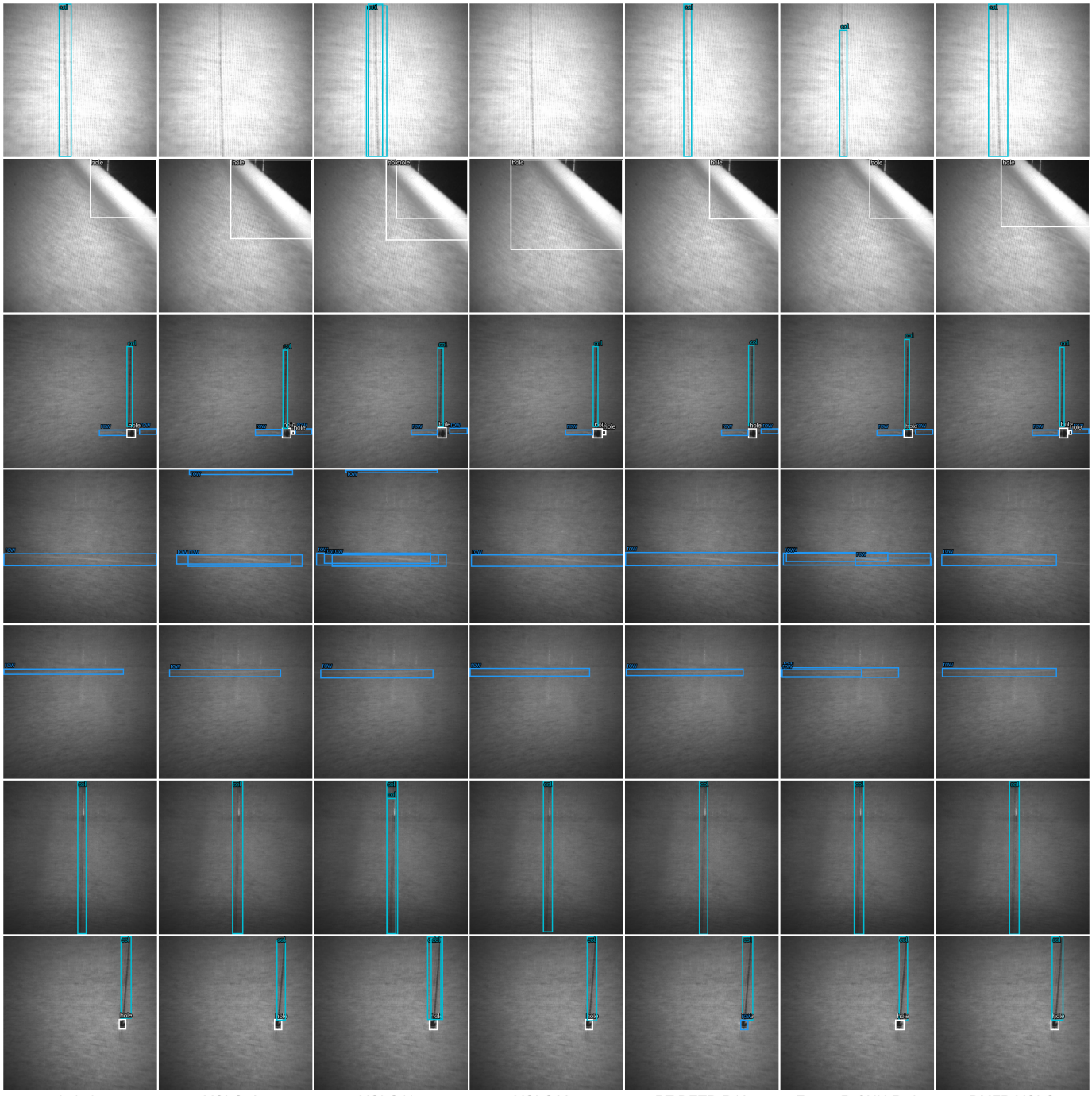

Fig. 7. Qualitative comparison of detection results on selected test images. From left to right, the columns show ground-truth annotations, YOLOv8n, YOLO11n, YOLO26n, RT-DETR-R18, Faster R-CNN-R50, and DMFR-YOLO (Ours).

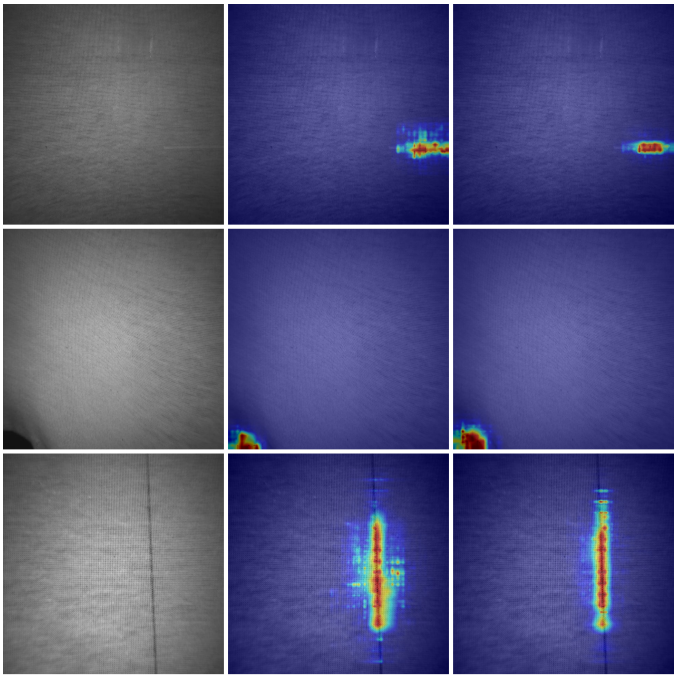

Fig. 8. Feature visualization of YOLOv8n and DMFR-YOLO. From left to right: original image (Origin), YOLOv8n, and DMFR-YOLO.

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